Processing and Analysis of Biological Data

Chapter 9. Causal inference

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Cause and correlation in biology

Correlation does not imply causation. This statement is central to scientific thinking, and underscores the importance of interpreting results from observational studies carefully, and ideally confirming any inferred relationship experimentally. Experiments are indeed a powerful way of separating the effects of multiple correlated variables. In this chapter, we will discuss an alternative approach to inferring causality tracing back to the work of Sewall Wright a hundred years ago (Wright 1921 and later). Broadly speaking, the method can be used to infer causality by combining knowledge about the natural history/mechanics of the study system with estimated statistical parameters such as correlation coefficients and regression slopes.

For further reading I strongly recommend Bill Shipley´s book "Cause and Correlation in Biology".

As an example, we will work with the alpine plants dataset.

```
plants = read.csv(file="datasets/alpineplants.csv")
```

Wrightian Path analysis

In its simplest form, a path analysis consists of a series of correlations combined with linear regressions fitted to standardized variables (zero mean, unit variance), thus obtaining *path coefficients*. Before going into technical aspects, a critical point is that before estimating any parameters, causal inference through path analysis or related methods requires formulating a graphical model in the form of a *directed graph* showing the assumed causal (and non-causal) relationships between a set of variables.

As an example, we will consider two different models for how snow depth, minimum winter soil temperature and growing-season soil moisture affect the distribution and abundance of the sedge *Carex bigelowii* (Cyperaceae). In the first model, we will assume independent effects of each predictor, thus building a path model on the form

$$snow \rightarrow Carex.bigelowii$$

$$min.T.winter \rightarrow Carex.bigelowii$$

$$soil.moist \rightarrow Carex.bigelowii$$

An alternative model is that snow cover affects winter soil temperature and growing-season soil moisture, which is turn affects the plant.

$$snow \rightarrow soil.moist$$

$$snow \rightarrow min.T.winter$$

$$min.T.winter \rightarrow Carex.bigelowii$$

$$soil.moist \rightarrow Carex.bigelowii$$

In path analysis, we call the response variables (with arrows coming into them) *endogeneous* variables, and the predictors (with arrows only going out of them) *exogeneous* variables.

The first model can be fitted as a standard multiple regression, while the second model will involve fitting three different component models. Before fitting the models, we remove some `NA`s and z-transform all variables (including the response variables). Removing `NA`s is important to make sure that our alternative models are fitted to exactly the same observations, and the z-transformation places all variables on a variance-standardized scale that will yield the path coefficients.

```
plants = na.omit(plants)
plants = as.data.frame(scale(plants))

round(colMeans(plants), 2)
```

```
##    Carex.bigelowii Thalictrum.alpinum      mean_T_winter       max_T_winter
##                  0                   0                  0                  0
##       min_T_winter      mean_T_summer       max_T_summer       min_T_summer
##                  0                   0                  0                  0
##              light               snow         soil_moist           altitude
##                  0                   0                  0                  0
```

```
round(apply(plants, 2, sd), 2)
```

```
##    Carex.bigelowii Thalictrum.alpinum      mean_T_winter       max_T_winter
##                  1                   1                  1                  1
##       min_T_winter      mean_T_summer       max_T_summer       min_T_summer
##                  1                   1                  1                  1
##              light               snow         soil_moist           altitude
##                  1                   1                  1                  1
```

```
m1 = lm(Carex.bigelowii ~ snow + min_T_winter + soil_moist, data=plants)

m2a = lm(min_T_winter ~ snow, data=plants)
m2b = lm(soil_moist ~ snow, data=plants)
m2c = lm(Carex.bigelowii ~ min_T_winter + soil_moist, data=plants)
```

```
summary(m1)
```

```
##
## Call:
## lm(formula = Carex.bigelowii ~ snow + min_T_winter + soil_moist,
##     data = plants)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.3948 -0.4935 -0.2902  0.2450  3.7531
##
## Coefficients:
```

```
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)  4.996e-16  9.775e-02   0.000    1.000
## snow         1.772e-01  1.629e-01   1.088    0.280
## min_T_winter 2.065e-01  1.647e-01   1.254    0.213
## soil_moist   2.254e-02  1.120e-01   0.201    0.841
##
## Residual standard error: 0.9427 on 89 degrees of freedom
## Multiple R-squared:  0.1404, Adjusted R-squared:  0.1114
## F-statistic: 4.844 on 3 and 89 DF,  p-value: 0.003612
```

The first model suggests positive but weakly supported effects of both snow cover and minimum winter temperature on the abundance of *Carex bigelowii*. Keep in mind though that snow cover and minimum winter soil temperature are strongly positively correlated, so that we may have some issues with multicollinearity in this model.

EXERCISE: Draw (on paper) the path diagram corresponding to this model, and add the estimated path coefficients, including the (Pearson) correlations (function `cor` in `R`) between the exogeneous (predictor) variables. We can calculate the unexplained variance ("U") in the response as $\sqrt{(1-r^2)}$ (which places it on the standardized [correlation] scale like the path coefficients). Interpret the results.

In this model we can calculate the total (net) effect of snow cover on the abundance of *Carex bigelowii* by summing the direct effect and the effects arising through correlations with other variables.

```
summary(m1)$coef[2,1] +
summary(m1)$coef[3,1]*cor(plants$snow, plants$min_T_winter, "pairwise") +
summary(m1)$coef[4,1]*cor(plants$snow, plants$soil_moist, "pairwise")
```

```
## [1] 0.3508336
```

```
cor(plants$snow, plants$Carex.bigelowii, "pairwise")
```

```
## [1] 0.3508336
```

In the second model, there is (as expected) a strong positive effect of snow cover on minimum winter soil temperature, and in turn a positive effect of winter soil temperature on *Carex bigelowii*. Thus, under this model, we have strong support for the hypothesized causal links from snow cover to *Carex* abundance.

EXERCISE: Draw the path diagram and interpret the direct and indirect effects of snow cover on *Carex* abundance.

```
summary(m2a)$coef
```

```
##                  Estimate Std. Error       t value     Pr(>|t|)
## (Intercept) -1.106648e-15 0.06354811 -1.741434e-14 1.000000e+00
## snow         7.927891e-01 0.06389254  1.240816e+01 2.829427e-21
```

```
summary(m2b)$coef
```

```
##                 Estimate Std. Error      t value     Pr(>|t|)
## (Intercept) 6.074519e-17 0.09345465 6.499965e-16 1.000000e+00
## snow        4.433825e-01 0.09396118 4.718784e+00 8.546112e-06
```

```
summary(m2c)$coef
```

```
##                  Estimate Std. Error      t value    Pr(>|t|)
## (Intercept)  6.733193e-16 0.09784898 6.881209e-15 1.000000000
## min_T_winter 3.389064e-01 0.11095020 3.054582e+00 0.002964704
## soil_moist   3.985433e-02 0.11095020 3.592092e-01 0.720279994
```

Structural equation modelling

Structural equation modelling is a further development that offers greater flexibility compared to traditional path analysis. Traditional structural equation models are estimated globally (in one go) based on a covariance matrix including all the candidate variables. An alternative approach is to build the SEM piecewise as a series of models fitted independently ("local estimation") and then combined. Below we fit our second candidate model using the second approach and the `piecewiseSEM` package.

```
library(piecewiseSEM)

m2 = psem(lm(soil_moist~snow, data=plants),
          lm(min_T_winter~snow, data=plants),
          lm(Carex.bigelowii~min_T_winter+soil_moist, data=plants),
          data=plants)
```

```
summary(m2)
```

```
##   |                                                                      |
```

```
##
## Structural Equation Model of m2
##
## Call:
##   soil_moist ~ snow
##   min_T_winter ~ snow
##   Carex.bigelowii ~ min_T_winter + soil_moist
##
##     AIC
##  683.464
##
## ---
## Tests of directed separation:
##
##                     Independ.Claim Test.Type DF Crit.Value P.Value
##      Carex.bigelowii ~ snow + ...      coef 89     1.0875  0.2797
##   min_T_winter ~ soil_moist + ...      coef 90     1.9656  0.0524
##
## --
## Global goodness-of-fit:
##
## Chi-Squared = 5.137 with P-value = 0.077 and on 2 degrees of freedom
## Fisher's C = 8.445 with P-value = 0.077 and on 4 degrees of freedom
##
## ---
```

```
## Coefficients:
##
##          Response    Predictor Estimate Std.Error DF Crit.Value P.Value
##        soil_moist         snow   0.4434    0.0940 91     4.7188  0.0000
##      min_T_winter         snow   0.7928    0.0639 91    12.4082  0.0000
##   Carex.bigelowii min_T_winter   0.3389    0.1110 90     3.0546  0.0030
##   Carex.bigelowii   soil_moist   0.0399    0.1110 90     0.3592  0.7203
##   Std.Estimate
##         0.4434 ***
##         0.7928 ***
##         0.3389  **
##         0.0399
##
##   Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05
##
## ---
## Individual R-squared:
##
##          Response method R.squared
##        soil_moist   none      0.20
##      min_T_winter   none      0.63
##   Carex.bigelowii   none      0.13
```

```
plot(m2)
```

Beyond simply fitting the component models, the `piecewiseSEM` package implements tests of so-called *directed separation* ("d-separation"), which is a test for conditional independence of variables. Here, the key test is if *Carex* abundance is really conditionally independent of snow cover, i.e. after we have accounted for winter soil temperature and growing-season soil moisture. The interpretation of these tests is different from what we are used to for normal models. In this case the null hypothesis is that our model represents the data well, and that adding an additional link (such as the one from snow cover directly to *Carex* abundance) would not improve the model sufficiently to be favoured. Therefore, statistical support (a low p-value or support in AIC comparison) therefore indicates *bad* model fit, while lack of statistical support (e.g. a higher p-value) indicate decent fit to the data. However, as always, we need to interpret any result in light of the parameter estimates.

Note that for a SEM containing multiple of these "independence claims", the `piecewiseSEM` package also provides a hypothesis test for the entire model, again with a high p-value indicating a decent fit to the data. This test is based on comparing the likelihood of the current model to a saturated model including the missing paths. The output of the current model includes a test of directed separation for the relationship between winter temperature and soil moisture, which was not our main interest in this case, and we would like to treat this as an untested correlation (corresponding to double-headed arrows in our traditional path analysis above). We can achieve this through the somewhat exotic `%~~%` operator within the `psem` function.

```
m2b = psem(lm(soil_moist~snow, data=plants),
          lm(min_T_winter~snow, data=plants),
          lm(Carex.bigelowii~min_T_winter+soil_moist, data=plants),
          min_T_winter %~~% soil_moist,
          data=plants)
summary(m2b)
```

```
##   |                                                                      |
```

```
##
```

```
## Structural Equation Model of m2b
##
## Call:
##   soil_moist ~ snow
##   min_T_winter ~ snow
##   Carex.bigelowii ~ min_T_winter + soil_moist
##   min_T_winter ~~ soil_moist
##
##     AIC
##  683.464
##
## ---
## Tests of directed separation:
##
##                 Independ.Claim Test.Type DF Crit.Value P.Value
##   Carex.bigelowii ~ snow + ...      coef 89     1.0875  0.2797
##
## --
## Global goodness-of-fit:
##
## Chi-Squared = 1.228 with P-value = 0.268 and on 1 degrees of freedom
## Fisher's C = 2.548 with P-value = 0.28 and on 2 degrees of freedom
##
## ---
## Coefficients:
##
##          Response    Predictor Estimate Std.Error DF Crit.Value P.Value
##        soil_moist         snow   0.4434     0.094 91     4.7188  0.0000
##      min_T_winter         snow   0.7928    0.0639 91    12.4082  0.0000
##   Carex.bigelowii min_T_winter   0.3389     0.111 90     3.0546  0.0030
##   Carex.bigelowii   soil_moist   0.0399     0.111 90     0.3592  0.7203
##    ~~min_T_winter ~~soil_moist   0.2029         - 93     1.9656  0.0262
##   Std.Estimate
##         0.4434 ***
##         0.7928 ***
##         0.3389  **
##         0.0399
##         0.2029   *
##
##   Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05
##
## ---
## Individual R-squared:
##
##          Response method R.squared
##        soil_moist   none      0.20
##      min_T_winter   none      0.63
##   Carex.bigelowii   none      0.13
```

AIC model selection for structural equation models

The AIC values reported by the model summary is the total AIC of the model, obtained by summing the AIC values of the component models (log likelihoods and thus AIC values are additive). We can use this

to compare our chosen model to a null model (with nothing affecting nothing), or an alternative model with different sets of paths. As an example, we formulate a third alternative model with different paths missing and compare it to our model 2 using AIC. Specifically, we omit the arrow from soil moisture to *Carex* abundance.

```
m3 = psem(lm(soil_moist~snow, data=plants),
          lm(min_T_winter~snow, data=plants),
          lm(Carex.bigelowii~min_T_winter, data=plants),
          min_T_winter %~~% soil_moist,
          data=plants)
summary(m3)
```

```
##   |                                                                      |
```

```
##
## Structural Equation Model of m3
##
## Call:
##   soil_moist ~ snow
##   min_T_winter ~ snow
##   Carex.bigelowii ~ min_T_winter
##   min_T_winter ~~ soil_moist
##
##     AIC
##  681.598
##
## ---
## Tests of directed separation:
##
##                       Independ.Claim Test.Type DF Crit.Value P.Value
##         Carex.bigelowii ~ snow + ...      coef 90     1.1337  0.2600
##   Carex.bigelowii ~ soil_moist + ...      coef 89     0.2013  0.8409
##
## --
## Global goodness-of-fit:
##
## Chi-Squared = 1.361 with P-value = 0.506 and on 2 degrees of freedom
## Fisher's C = 3.041 with P-value = 0.551 and on 4 degrees of freedom
##
## ---
## Coefficients:
##
##          Response    Predictor Estimate Std.Error DF Crit.Value P.Value
##        soil_moist         snow   0.4434     0.094 91     4.7188  0.0000
##      min_T_winter         snow   0.7928    0.0639 91    12.4082  0.0000
##   Carex.bigelowii min_T_winter   0.3573    0.0979 91     3.6497  0.0004
##    ~~min_T_winter ~~soil_moist   0.2029         - 93     1.9656  0.0262
##   Std.Estimate
##         0.4434 ***
##         0.7928 ***
##         0.3573 ***
##         0.2029   *
##
```

```
##   Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05
##
## ---
## Individual R-squared:
##
##          Response method R.squared
##        soil_moist   none      0.20
##      min_T_winter   none      0.63
##   Carex.bigelowii   none      0.13
```

```
AIC(m2b, m3)
```

```
##       AIC  K  n
## 1 683.464 10 93
## 2 681.598  9 93
```

The second model (`m3`) is favoured because it has the lowest AIC value.

One nice feature of the `piecewiceSEM` package is that is allows the component models to be for example GLM's or mixed models. This offers greater flexibility for including non-Gaussian variables, and for modelling structure in the data (e.g. non-independence of data points).

EXERCISE: Repeat the analyses above for *Thalictrum alpinum* instead of *Carex*.

EXERCISE: Can you think of other potential models that can be tested? Are there other important environmental variables? Start by drawing the competing models as directed graphs (on paper). Fit and compare the models, and interpret the results.